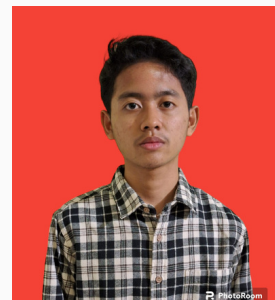


Sulthon Akbar Robbani

082229693458 | benyyy990@gmail.com | www.linkedin.com/in/sulthon-akbar

Undergraduate Mechanical Engineering Student at Bandung Institute of Technology

Motivated 6th-semester Mechanical Engineering student with experience in mechanical design, manufacturing, and organizational leadership, highly interested in developing my skills in the energy transition and renewable energy sector, as well as in manufacturing



EDUCATION

Institut Teknologi Bandung – Bandung, Jawa Barat

Aug 2023 – Present

Bachelor of Mechanical Engineering,

3,59/4,00

PROJECT EXPERIENCE

Clean water supply project in Kabetan village, Central Sulawesi with solar panel pumps

Apr 2026 – Present

- **System Design & Sizing :** Designed a 24V off-grid solar-powered water pumping system with a 280W submersible pump, including technical sizing of a 120Ah GEL battery at a 50% Depth of Discharge (DoD) to ensure a stable 5-hour daily operational target.
- **Electrical Protection:** Implemented a Dual-Layer Overcurrent Protection system using 16A DC MCBs positioned at the PV-to-SCC and SCC-to-Battery lines to safeguard components against thermal runaway and voltage surges.
- **Automation & Controls:** Developed a smart hybrid water level management system combining an electrical sensor at the upper reservoir (Toren) and a mechanical float valve at the village distribution tank to prevent water overflow and optimize energy storage.
- **Hydraulic & Filtration Engineering:** Configured a high-flow filtration line using a 10-inch Nanotec sediment filter (1" thread) placed sequentially after a brass check valve to prevent water hammer, complemented by a dual-valve hydraulic bypass system to facilitate zero-downtime maintenance.

ENGINEERING EXPERIENCE

Aksantara ITB 2025/2026

Staff of Fixed Wing Manufacturing Department

Mar 2025 – Dec 2025

- fabricated fuselage molds
- conducted composite lay-up and vacuum bagging processes
- performed structural repairs on damaged aircraft used for flight testing before the KRTI competition.

COMMITTEE EXPERIENCE

Mechanical Festival ITB 2026

Vice Head of Public Relation Division

Oct 2025 – Present

- Disseminated MFEST information to high schools and universities to increase awareness and participation.
- Acted as a communication liaison between MFEST and external stakeholders, including sponsors and speakers.
- Coordinated interactions with alumni and companies to support event engagement and collaboration.

Wisuda Oktober HMM ITB 2025

Liaison Officer

Sep 2025 – Nov 2025

- Acted as a liaison between graduates and the graduation committee to facilitate communication.
- Disseminated important graduation-related information to all graduates.
- Ensured that graduates' needs and requirements were met throughout the event.

Wisuda April HMM ITB 2025

Vice Head of "Penguasanaan" Division

Mar 2025 – Apr 2025

- Designed and produced creative props to enhance the graduation parade experience.
- Planned and coordinated the execution of the commemorative performance.
- Delivered a final tribute performance to honor and celebrate graduating students.

Aksi Angkatan Mesin 23 ITB

Head of Research and Development Division

Sep 2024 – Nov 2024

- Analyzed the needs of Cupunagara Village by comparing actual conditions with ideal benchmarks.
- Collected, processed, and analyzed data from village surveys to support evidence-based planning.
- Developed community programs based on survey data analysis and the gap between actual and ideal conditions.

Parade Wisuda Oktober ITB 2023

Field Staff

Aug 2023 – Oct 2023

- Coordinated participant flow to ensure strict compliance with directives during the graduation parade.
- Maintained the integrity of concurrent events by overseeing smooth operations.
- Minimized external distractions to ensure uninterrupted and orderly event execution.

Gathering FTMD ITB '23

Field Coordinator

Jul 2023 – Aug 2023

- Managed technical operations and overall event execution on the event day.
- Coordinated the mobilization and movement of 360 FTMD ITB Batch of 2023 participants.
- Ensured orderly and timely participation to support smooth event flow.

ORGANIZATIONAL EXPERIENCE

Badan Pengurus HMM ITB 2025/2026

Staff of Appreciation Division

Sep 2025 – Present

- Collaborated with team members to collect and organize data on members' birthdays.
- Managed records of championship and competition winners, as well as graduation data.
- Supported recognition and appreciation initiatives aimed at acknowledging all members of HMM ITB

Badan Pengurus Angkatan 23 Mesin ITB

Staff of Internal Relation Division

Mar 2024 – Present

- Organized activities to strengthen solidarity among batch members.
- Facilitated events aimed at fostering a supportive and cohesive batch atmosphere.
- Promoted engagement and collaboration within the batch to enhance overall group unity.

Badan Pengurus HMM ITB 2024/2025

Intern at Alumni Relation Division

Jan 2025 – Jun 2025

- conducted alumni interviews
- developed digital magazine content for Instagram,
- facilitated communication between alumni and HMM ITB.

ACADEMIC PROJECT

Gearbox Design and Manufacturing for Power Transmission

Mei 2025 – Dec 2025

- Analysis and Design: Conducted force and motion analysis, calculated safety factors, determined tolerances, and performed dimensional analysis according to ISO standards for components including shaft, housing, base plate, bearings, gears, keys, and keyways.
- Technical Drawing: Developed detailed technical drawings and 3D models using SolidWorks for all gearbox components.
- Manufacturing and Assembly: Executed the manufacturing process using lathe, milling, drilling, and wire-cut machines; performed assembly of gearbox components.
- Testing and Measurement: Conducted comprehensive testing and measurements to verify compliance with design specifications, including: Coordinate Measuring Machine (CMM) for complex geometries and total runout tolerances, Surface roughness tester, Profile projector, and Direct measurement tools such as vernier calipers, micrometers, and the three-wire method

TECHNICAL SKILLS AND TOOLS

- Mechanical Design and Technical Drawing : Basic SolidWorks
- Manufacturing : Lathe, Milling, and Drilling Basic Operation
- Piping System Design : Basic AutoPipe
- Other : Basic Microsoft Word, Excel, Power Point, and Canva

BASIC ENGINEERING SKILLS

- Basic Structural, Force, and Motion Analysis – Conducted fundamental analysis of forces, motions, and structural behavior in mechanical systems.
- Basic Mechanical Vibration Analysis – Performed preliminary assessments of vibration characteristics in mechanical components.
- Basic Design Analysis – Applied fundamental principles of tolerance design and safety factor calculation to ensure component reliability.
- Basic Stress and Fluid Flow Analysis – Performed fundamental stress analysis and pipe fluid flow calculations to support safe and efficient mechanical system design
- Basic Electrical and Power System Assembly – Built and analyzed basic electrical circuits using DC motors, single-phase and three-phase AC motors, capacitors, diodes, amplifiers, and power generators for fundamental engineering applications.
- Basic Hydraulic and Pneumatic System Understanding – Applied fundamental principles of hydraulic pumps and pneumatic systems to support basic fluid power operation and system analysis.
- Basic Vibration Analysis – Conducted fundamental analysis of free and forced vibration to understand dynamic behavior in mechanical systems.
- Basic Sensor Analysis – Evaluated the working principles and applications of temperature, speed, and position sensors for basic instrumentation and control systems.

SOFT SKILL

- Effective Communication, Leadership, Adaptive Learning, Time Management, problem solving, people management, and decision making

AWARDS AND SCHOLARSHIPS

- Awardee of Beasiswa Indonesia Bangkit (2023–Present)